

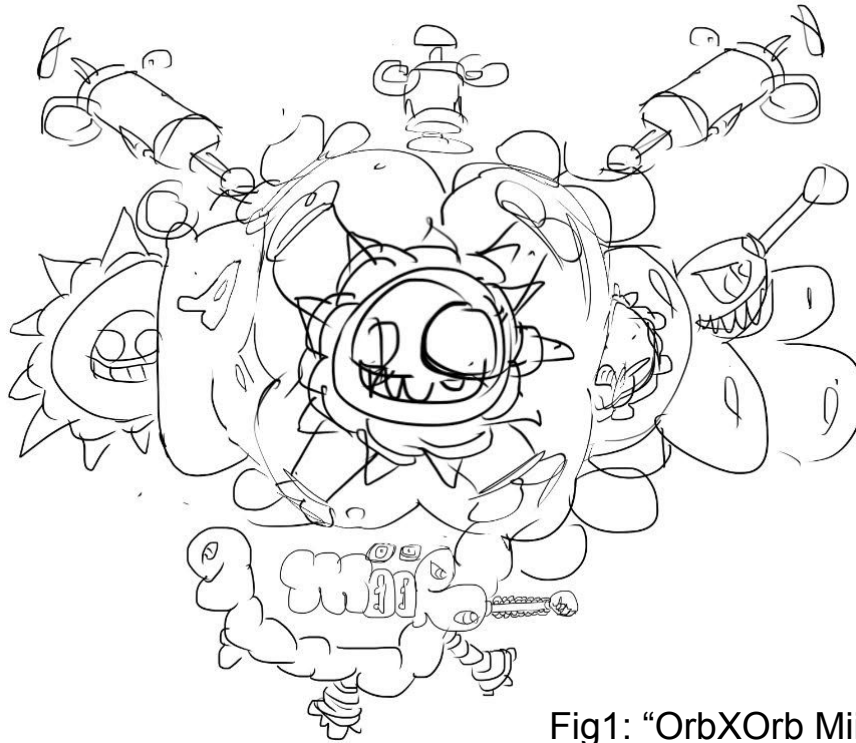
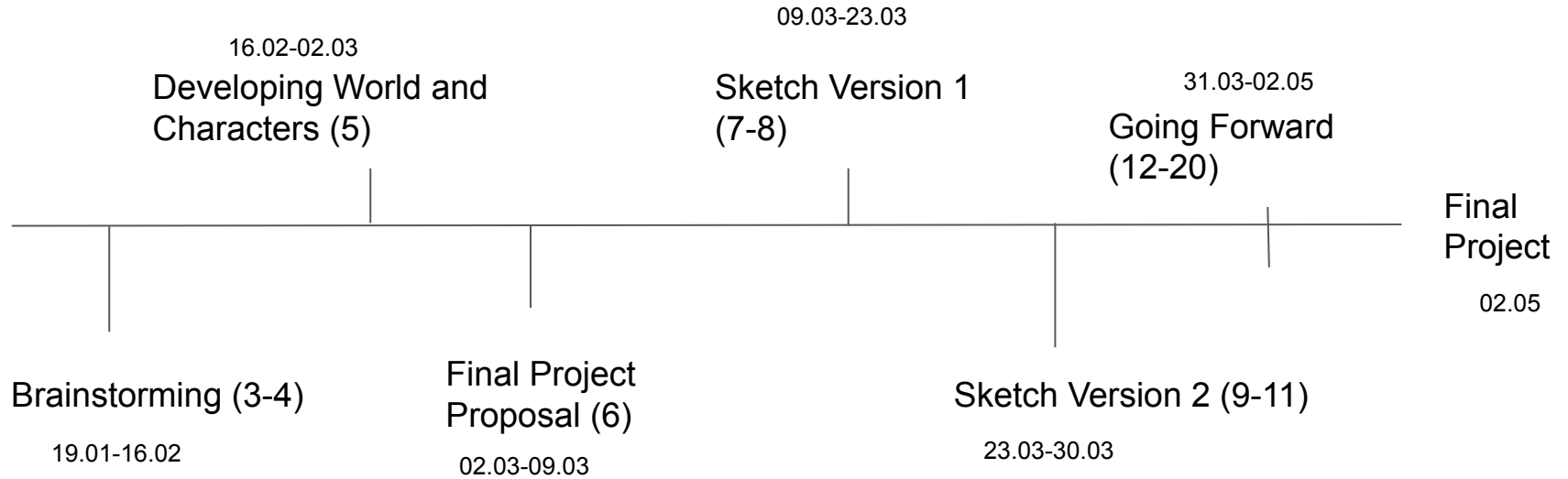


Fig1: “OrbXOrb MiiR”

OrbXOrb MiiR: An interactive virtual-physical art exhibition

Timeline



Inspiration

This project was inspired an exhibition by LuYang while visiting back home in Basel. I was mesmerised by the virtual world Yuang created and gave the ability to interact with via a game controller (Fig3). This fit the theme of inside/outside as it mapped the users input from the outside and interacted with the virtual world. I decided to do something similar but using my own practise as a basis for it. I would create a virtual world from my frequently drawn characters being set with maximalist and toy like theme, it would explore the world of my imagination.

Artists such as SerotinesArt inspired me to adapt my art into 3D, as Ive found the way the way they bring drawings to life perfect for showcasing my are in a new dimension (Fig2).

Walter van Beirendonck, characters and universe surrounding his namesake label, Wild & Lethal Trash and specifically the ikea collaboration Glödande inspired both my artstyle and world building as he is one of my favourite designers (Fig4). For Ikea's Glödande Walter create a fictional universe in the clouds which inspired me to create my own universe constrained to sphere of my drawings.

The name OrbXOrb MiiR comes from my tattoo name OrbXOrb and MiiR comes from the russian word for world.



Fig2: serotinesart (2023)



Fig3: Yang, Kunsthalle Basel and Hänger, (2023)



Fig4: Van Beirendonck and IKEA
(2016)

Poster Plan

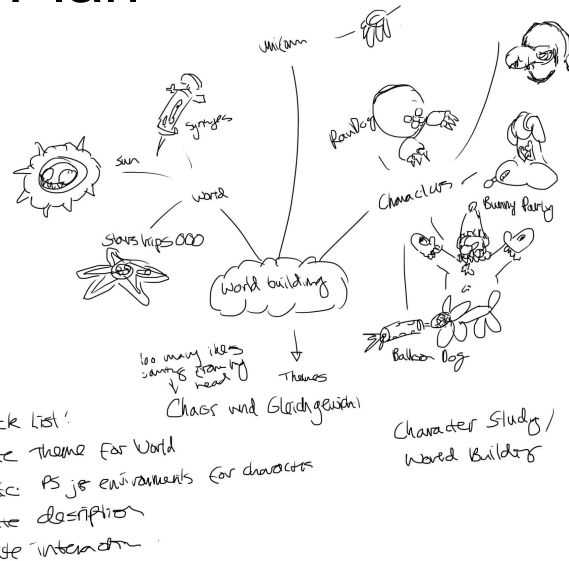
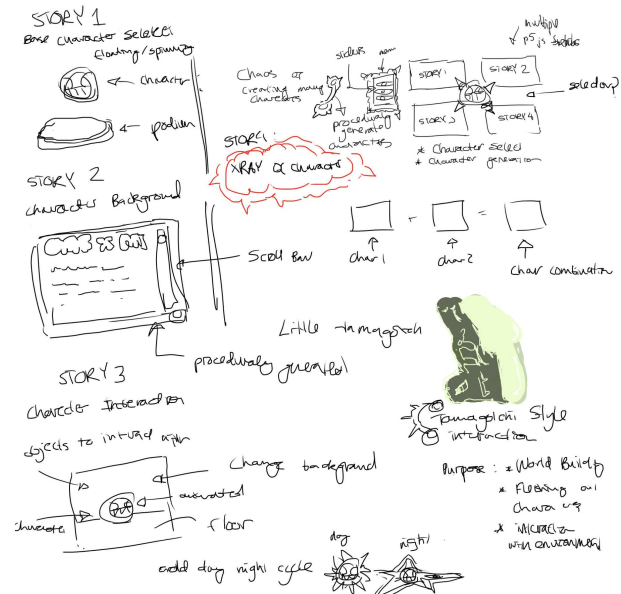


Fig5: World building mind map

In the brainstorm above I explored world building with the characters I draw, thinking about the possible themes such as the chaos and maximalism of my work. I created a checklist of things I would need to do in following sessions for the project (Fig5).

At first my project idea was based around world building and creating a showcase of my characters (fig6). However I decided to focus on showcasing my characters and their interactions by creating a virtual gallery where you are able to explore the world with the illustrations and 3D modelled characters.

Fig6: Brainstorming p5.js sketches



World and Characters

I created the World using a sphere and a texture of my favourite sketches. This is a good representation of my art and an appropriate backdrop to theme the gallery. I created the texture in ibis paint and exported it as a png to p5js (Fig7).

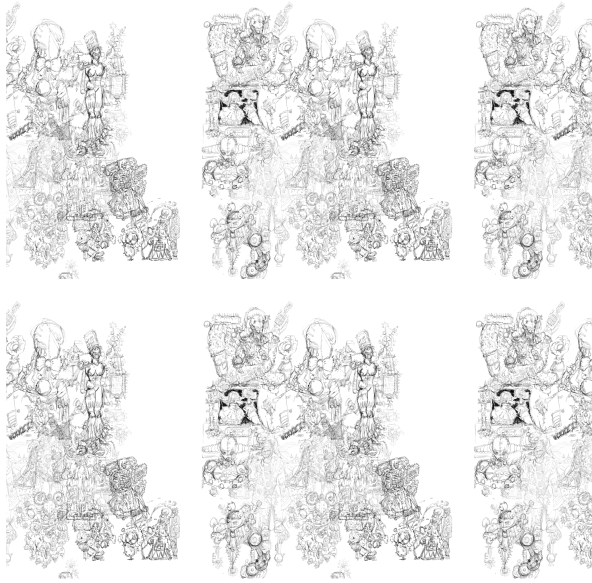


Fig7: Background created in Ibis paint



Fig8: Face

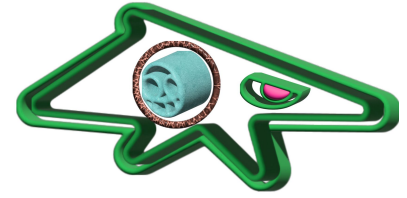


Fig9: Star3D

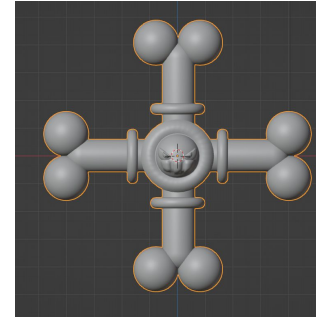


Fig10: Plus

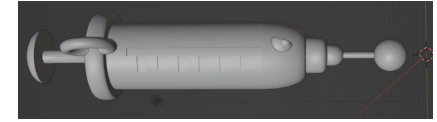


Fig10: Syringe Ship

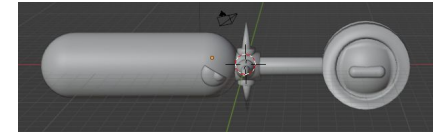


Fig11: Minus

The character and art pieces I created using different digital mediums. Some were create in adobe illustrator from my ibis paint sketches (Fig8-9). For this project I tried learning blender to expand my current skill set and created some character models within my hardware limits (Fig10-11). I will try to create more models of the characters I had in my poster plan as I am thinking about getting a more powerful computer for blender.

Proposal

Goal: Create a virtual exhibition of my current artwork and allow it to interact with the digital and physical world.

Functionality: The main functionality of the poster will be the ability to explore the different artworks and sculptures within the OrbXOrb MiiR (world). I am planning on adding interactions such as projectiles to fire at the artworks/sculptures and get information to pop up about them and play sounds based on my feedback I was also advised to make the artworks collectable which I might try to implement.

The second part will be the AR portion of the project which will be triggered when the player escapes the world boundaries. This will activate a new javascript file which will allow the user to use their phone camera and a QR code on the screen to display the 3D syringe models and interact with the physical environment. The interaction will come from being able to fire projectiles and create virtual objects within the AR space such as the sculpture and drawings.

I believe that overall these two dynamics will effectively tackle the mapping and inside-outside theme through the juxtaposition of digital-physical worlds, world exploration, 3D-2D worlds and world Boundaries. (a more detailed description of the functionality can be found on the poster itself.)

Audience: I will likely upload my final project as a link on my tattoo page. I believe it would be a good showcase of my current art pieces and portfolio in essence marketing my work. My target demographic would be ages 18-27 within the London area interested in tattoos and digital art. The format of the poster is appropriate as it will be made accessible from any laptop/pc specifications

Inspiration:

- LuYang: Digital Physical Interaction (Yang, Kunsthalle Basel and Hänger, 2023)
- Walter Van Beirendonck: World Building (Van Beirendonck and IKEA (2016))
- Serotines: 3D Models (serotinesart ,2023)

Themes:

- Art: Toys, Maximalism, Oddities, Chaos
- Mapping: digital-physical worlds, world exploration, 3D-2D worlds.
- Inside-Outside: digital-physical, World boundaries

Poster Version 1 - IN



3D syringe model that orbits around the scene. So far the only interaction implemented is orbiting around the syringe to view the drawing within the scene. I am planning on adding more interaction and animation to the syringe such as textures and ability to fire projectiles and a propulsion jet.

Everything is kept to my Arts themes such as toys maximalism and oddities, to give coherent dive into my art style

Play story button allows the description to be read out by a narrator giving a story world sort of feel

Selector to change between the inside and outside dimension. Inside dimension has the texture of my artworks while the outside uses the camera to map a video feed texture

Poster Version 1 - OUT

Same layout as in
the in dimension



Sphere now has the texture of the camera emulating the way the AR implementation would work. For the poster I decided to focus on creating a series of characters and showcasing them as examples of what I will populate the world with. For the final project I will be adding more interaction as well as functionality such as phone based AR. Once the user flies out of the world boundary a QR code will appear as well as a link allowing them to use their mobile phone as the camera and the QR code as the place for the 3D model to show. This will add more interaction between the viewer and the physical world.

Version 2 will have the project description for the
outside dimension

Iteration based on feedback

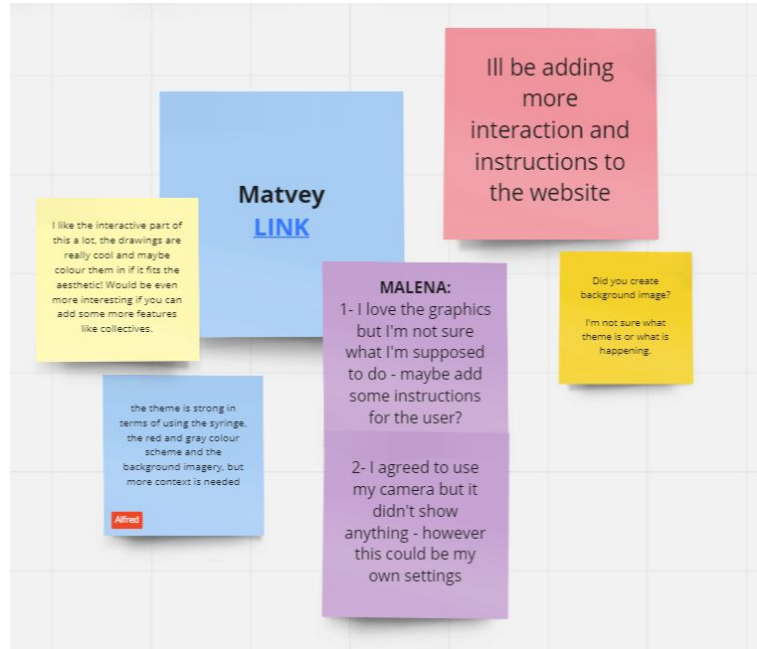


Fig12: Feedback

Adding collectives - from what I understood this would be adding collectables to the game. This would be an interesting concept such as having the AR world be triggered when enough artworks have been collected. This would also incentivise the user to explore the world. I am thinking about implementing this.

More Context (Theme) - Unfortunately in the first design I forgot to include the description of the project. I will make sure in the next design to include it. Although people did understand the cohesiveness of the artstyle leading to an exploration of my art style.

Instructions - Similar to more context I will add instructions on the next poster explaining the purpose of the project. However in the final project I will make the controls more intuitive such as the classic WASD movement controls and mouse camera controls for exploration.

Camera Functionality - We later figured out that was a client side issue which unfortunately I was not able to solve. Hopefully the AR version will be more accessible.

Summary

From the feedback I will add a description and instructions for the user in the final poster, however for the final project I will make it less guided to add an exploration element. I will also be adding collectables as it would add incentive for the user to explore the space.

Poster V2 - IN

Update the background with more recent graphics

Added new artworks to be displayed in the center of the sketch as the syringe orbits around. These will be dispersed throughout the world and be able to be interacted with by firing a projectile at them to receive information on them.



Added new selector for the artwork in the center, in the final project, which also narrates what object is in the center using text to speech upon switching

Added a description based on feedback for the inside portion of the poster

Poster V2 - OUT

Outside portion remained generally the same just with the new updates as for the inside dimension



The 2D artworks are able to be displayed as textures on a variety of shapes

Going Forward

Goals going forward will be to implement sound design such as the atmosphere and when object interact.

A big one will be actually adding the AR capabilities but after doing some research I found that there are javascript libraries capable of doing so.

In order to make my sketch more dynamic I still need to add more Interactions and Animation. A primary interaction I want to implement is to be able to fire projectiles at object to view information on them and be able to collect them. I also want to add small details such as propulsion for the rocket ship and some particle effects and animations for the artworks. Lastly adding exploration controls such WASD movement and camera mouse controls would finalize the exploration aspect of the game, as well as dispersing the characters within the world.

Also adding textures for the 3D models and more artworks would help flesh out the world. From fig5 I'm planning to add "RanDog" and "BalloonDog" to the world among others.

I believe I've set myself up in a good position with the poster to be able to develop it into the final project over the april break.

Checklist

- Sound Design
- AR capabilities
- Interaction + Animation
 - Projectiles
 - Propulsion
 - Artwork Particles and Effects
 - Exploration and Controls
 - Make Artwork Collectable
- Artworks:
 - 3D Textures
 - More Artworks

Going Forward - Final Project

During the easter break I worked on a variety of features from what I wanted to do going forward.

I was able to implement AR capabilities via a web based solution called mywebAR. Although this does not offer as much customizability in comparison coding your own AR solution, it was the most effective solution as my focus for the project shifted from the literal inside outside exploration but as a way to allow people not familiar with my art to be able to experience it in a dynamic and fun environment.

Therefore I focused primarily on world building for the syringe ship, adding new textures, an overlay, picture frames and particles and projectiles to have elements of my art embedded within all the structures. I also created 3rd person controls allowing the user to interact with the environment more easily compared to my prototype where the syringe would circle around the center allowing a minimal field of view. Since I wanted to focus on the actual interaction experience within the sketch, I decided to get rid of the voice over as I deemed it too distracting as well as keeping everything within the main screen offering a more cohesive experience.

Unfortunately I was not able to Sound Design nor special effects for each artwork, I focused a lot of my time on creating new models, textures and tweaking the gameplay experience to a point where my attempted sound use would have detracted from the experience as it would be too distracting. As an extension I would add more individual effects to each of the art pieces to make them feel even more unique, as well as adding a sort of card collection system.

Overall this was a very informative experience, where I was able to gather skills in 3D Modelling, Texturing and manipulating 3D space with ample opportunities to extend.

Checklist

- Sound Design
- ~~AR capabilities~~
- ~~Interaction + Animation~~
 - ~~Projectiles~~
 - ~~Propulsion~~
 - Artwork Particles and Effects
 - ~~Exploration and Controls~~
 - ~~Make Artwork Collectable~~
- ~~Artworks:~~
 - ~~3D Textures~~
 - ~~More Artworks~~

Going Forward - Developing Ideas

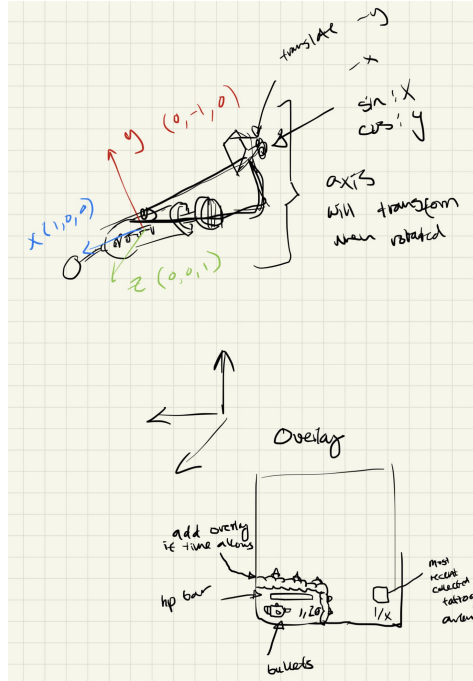


Fig13: Camera Controls and Overlay

After going in for easter break I started brainstorming on how to add collectables, movement and exploration to the game.

In fig13 I was exploring how I could get the camera to follow the ship while allowing to do mouse controls. As well as creating a prototype overlay.

In fig14 I was developing a restart screen allowing the user to reset the game once all the tattoos have been collected or they want to beat their previous time.

In fig15 I played around with designed a bullet for the ship to fire I decided on the design in the middle as it would be the most realistic with my current blender skill set.

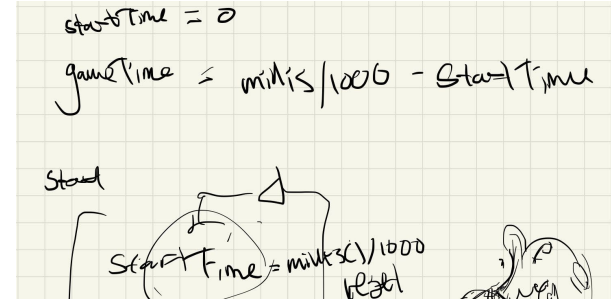


Fig14: Restart Development

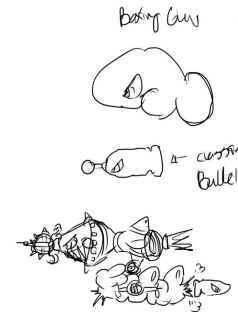


Fig15: Bullet Concept

Going Forward - 3D Models and Textures

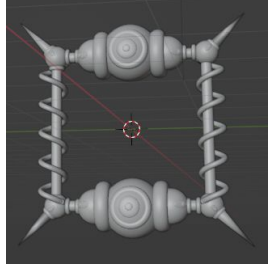


Fig16: Frame



Fig17: Textured Bullet

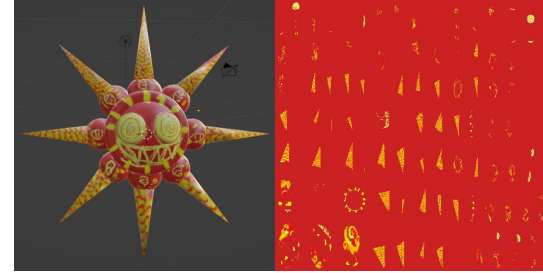


Fig18: Textured Sun Particle

Here are some examples of the new models and textures. I found it a particularly fun experience texturing although it was quite frustrating at the beginning as unless I was in a flow it was difficult to come up with ideas for the colors and the drawings. This exercise helped me to become more confident in blender and create models easier, although I definitely want to learn more about texturing and how I can combine it with different material properties.

The bullet (Fig17) was my first new model then I did the sun (fig18) however after completing the Frame (Fig16) I decided to slightly alter them with my new blender knowledge to make them more coherent

Going Forward - AR and Overlay



Fig19: AR portion

On the left I developed that AR portion of the sketch (Fig19). I used mywebAR where I uploaded a gltf file to be mapped onto the QR code. I believe mywebAR was the best solution as it was the simplest and most accessible.

I developed my initial overlay design from page 14 into a design (Fig 21) full of my drawings, as if they are spilling in from the background into the sketch. Although quite cluttered this adds to the chaotic element I was going for with this design.



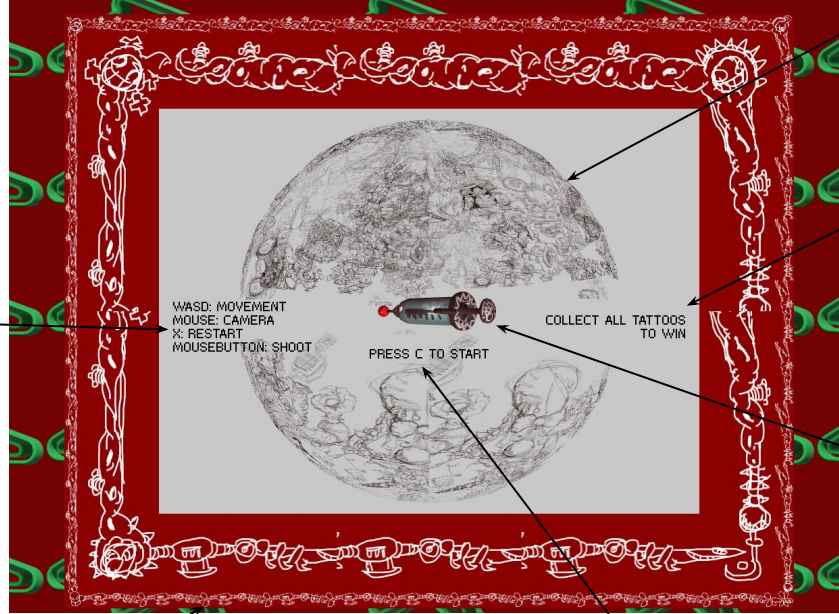
Fig20: no Overlay End Screen



Fig21: Overlay End Screen

Going Forward - Final - Start Screen

This is simple start screen for the project giving a glimpse of what's to come, and very simple breakdown of the controls a point I worked on from my feedback. As well as creating a goal to make it more understandable what you can actually do.



Teaser of world border

Goal of the game

Menu explaining Controls

Teaser of your game character rotating in a circle

No more description as a separate DIV, and text to speech as it would be distracting

Indication of how to start

Going Forward - Final - Game Screen

Although a bit cluttered, this was the look I was going for in the main part of the sketch. I wanted it to be loud and full of chaos and color reminiscent of my art style. The hud is quite intuitive and the controls are easy to pick up as they follow quite standard WASD and mouse movement controls. It's quite easy to navigate and explore the world offering the user an unforgettable experience.



Checks time elapsed after sketch has started. Resets upon restart

Overlay

Frame - Is drawn around the tattoo to give it more structure. Discovered tattoos have their description above them.

Artwork Hud - Tracks how many of the tattoos have been found and shows the most recently collected piece

Bullet Hud - Shows the bullets model that has a traffick light system telling you when you need to reload (get progressively more tinted red) and tracks the amount of bullets remaining

Particle Propulsion

Syringe Ship - the vessel you control using WASD and the mouse to change the camera View. Clicking fires a projectile that can be used to mark tattoos as collected. Going over / near a tattoo also sets it as collected

Going Forward - Final - End Screen

An easy to follow end screen giving the user to options of restarting or scanning the QR code making the navigation process smooth. The QR code is lit up with a special light as part of the overlay so it will be read more easily on the phone. The advantage of using a web based AR solution is also that it is more accessible.

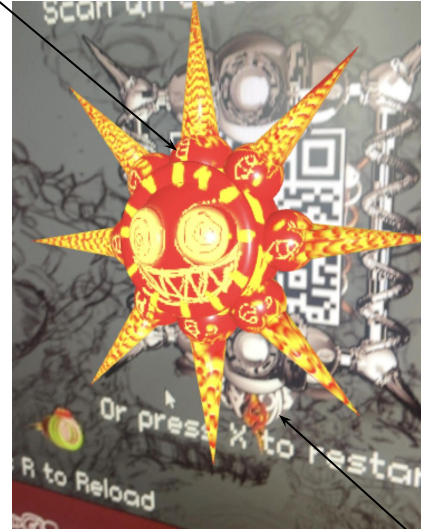
Instructions - make it clear for the user what options they have after completing the game

Overlay

Previously Mentioned Hud Elements



WebAR QR Code



AR Collectable - Although quite simple, adding the element of making the user get out their phone physically, scan the QR code and view the 3D model come to life, adds a new dimension to the project, almost like letting the world built within the sketch spill out into real life playing with the concept of inside and outside.

As a possible extension I would add more artworks that could be randomly collected and maybe make them procedurally generated

Updated Proposal

Changes are written in Bold

Goal: Create a virtual exhibition of my current artwork and allow it to interact with the digital and physical world. **The goal has remained mostly the same but without a heavy emphasis on interaction in the physical world.**

Functionality: The main functionality of the poster will be the ability to explore the different artworks and sculptures within the OrbXOrb MiiR (world). I am planning on adding interactions such as projectiles to fire at the artworks/sculptures and get information to pop up about them and ~~play sounds~~ based on my feedback I was also advised to make the artworks collectable which I might try to implement **which I implemented but I could add more elements such as collection cards.**

The significance of the AR portion has decreased however it still displays an AR collectable at the end of the game rather than once escaping the boundaries. Although the interactiveness of the AR portion decreased the focus on creating a virtual exhibition of my work increased which was the main idea for this project. Also by adding new models, textures and overlay I was able to add more material for the viewer to explore and become more acquainted with my art style.

I believe that overall these two dynamics will effectively tackle the mapping and inside-outside theme through the juxtaposition of digital-physical worlds, world exploration (**world exploration has become a lot more significant as its the primary focus with the new camera and movement controls**) and 3D-2D worlds and ~~world boundaries.~~

Audience: I will likely upload my final project as a link on my tattoo page. I believe it would be a good showcase of my current art pieces and portfolio in essence marketing my work. My target demographic would be ages 18-27 within the London area interested in tattoos and digital art. The format of the poster is appropriate as it will be made accessible from any laptop/pc specifications. **The audience has remained completely the same.**

Inspiration:

- LuYang: Digital Physical Interaction (Yang, Kunsthalle Basel and Hänger, 2023)
- Walter Van Beirendonck: World Building (Van Beirendonck and IKEA (2016))
- Serotines: 3D Models (serotinesart ,2023)

Themes:

- Art: Toys, Maximalism, Oddities, Chaos
- Mapping: digital-physical worlds, world exploration, 3D-2D worlds. **World exploration has been given greater significance.**
- Inside-Outside: digital-physical, ~~World boundaries~~

Overall I learned about the vast capabilities of p5js, blender and some game design with many capabilities to extend this project further.

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